

Why Rotating Fluids Make Polygons

Abstract

Saturn has a hexagon. Jupiter has an octagon and a pentagon. We show that polygonal vortex structures are inevitable in two-dimensional rotating fluids through a chain of four results: (1) The Green’s function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, which is monotonically decreasing; this forces the point vortex interaction to be attractive. (2) For any rotationally symmetric Hamiltonian with attractive pairwise interaction $h(r)$ satisfying $h'(r) < 0$, the regular N -gon maximizes the interaction energy along the spiral deformation direction; we prove $H''(0) < 0$ for all $N \geq 3$ via an explicit closed-form formula. (3) The Kraichnan inverse energy cascade in 2D turbulence drives energy to large scales while dissipating enstrophy at small scales, producing a negative-temperature state in the Onsager (1949) sense. (4) At negative temperature, the most probable macrostate maximizes energy—selecting the polygon.

Three planetary configurations are tested: Saturn’s hexagon ($N = 6$), Jupiter’s north polar octagon ($N = 8$), and south polar pentagon ($N = 5$). All three sit near the polygon locus despite arising through different dynamical mechanisms (Rossby wave stationarity for Saturn, discrete vortex self-organization for Jupiter). *Which* polygon forms depends on planetary parameters; *that* a polygon forms is universal—a consequence of two-dimensionality, symmetry, and statistical mechanics.

1 Introduction

In 1988, David Godfrey examined Voyager images of Saturn’s north pole and found a hexagon [Godfrey, 1988]—a persistent, regular, six-sided polygon inscribed in the jet stream at 76° N, stable across decades. Thirty years later, the Juno spacecraft revealed that Jupiter’s poles harbor their own polygonal arrangements: eight cyclones in an octagonal ring at the north pole, five in a pentagonal ring at the south [Adriani et al., 2018]. Laboratory rotating tanks produce similar patterns [Barbosa Aguiar et al., 2010].

These observations raise a question with two parts. The first—*how* does a specific polygon form?—has received considerable attention. Numerical simulations reproduce Saturn’s hexagon from jet instabilities [Morales-Juberías et al., 2011]; vortex dynamics models explain Jupiter’s cyclone numbers [Gavriel and Kaspi, 2021]; Rossby wave theory predicts the hexagonal wavenumber $n = 6$. The second part—*why* should different physical mechanisms on different planets all produce the same geometric outcome?—has not been addressed.

This paper answers the second question. The argument has four steps:

1. **The interaction is attractive.** Planetary atmospheres are two-dimensional surfaces. The Green’s function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, which is decreasing. This forces the vortex interaction $h(r)$ to satisfy $h'(r) < 0$.

2. **The polygon is an energy maximum along the spiral direction.** For N point vortices on a ring with any pairwise interaction satisfying $h'(r) < 0$, the regular N -gon is a critical point and local energy maximum along the centered spiral deformation: $H'(0) = 0$ and $H''(0) < 0$ for all $N \geq 3$ (Theorem 1). Whether it is a maximum in the full $(2N - 3)$ -dimensional configuration space is an open question.
3. **2D turbulence operates at negative temperature.** The Kraichnan (1967) inverse energy cascade transfers energy to large scales while the direct enstrophy cascade dissipates enstrophy at small scales. This separation produces a negative-temperature state in the Onsager (1949) sense.
4. **Negative temperature selects energy maxima.** At negative temperature, the Boltzmann weight $P \propto e^{-\beta H}$ with $\beta < 0$ assigns higher probability to higher-energy states. The polygon, being the energy maximum along the spiral direction, is a candidate for the most probable macrostate.

Steps 1 and 2 are proven within this paper. Step 3 is established fluid mechanics [Kraichnan, 1967]. Step 4 is established statistical mechanics [Onsager, 1949].

Which polygon forms ($N = 5, 6, 8$) depends on planet-specific mode selection: Rossby dispersion for Saturn, vortex dynamics for Jupiter. *That* a polygon forms is universal—a consequence of two-dimensionality, symmetry, and statistical mechanics.

2 Setup

2.1 Point vortex dynamics on a 2D surface

On a two-dimensional surface, the stream function ψ satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y, \quad (1)$$

where β is the meridional gradient of the Coriolis parameter. The Green's function of the 2D Laplacian is

$$G(r) = -\frac{1}{2\pi} \ln r, \quad (2)$$

which satisfies $G'(r) = -1/(2\pi r) < 0$ for all $r > 0$. This is not a modeling choice; it is forced by the ellipticity of ∇^2 in two dimensions.

For N point vortices with equal circulations κ , the interaction energy is

$$H = -\frac{\kappa^2}{4\pi} \sum_{j < k} \ln |z_j - z_k| = \frac{\kappa^2}{2} \sum_{j < k} G(|z_j - z_k|). \quad (3)$$

Since G is decreasing, the pairwise interaction is *attractive*: vortices at closer separation have lower energy.

2.2 The centered spiral deformation

Consider N vortices arranged on a ring of radius R , deformed along the centered one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (4)$$

At $\sigma = 0$: the regular N -gon. For $\sigma \neq 0$: each vertex is displaced radially by a different amount, spiraling the configuration. The centering $(k - (N - 1)/2)$ ensures that $\sigma = 0$ is a *critical point* of H : under the relabeling $k \mapsto N - 1 - k$, the exponent maps to $-(k - (N - 1)/2)/N$, sending $\sigma \rightarrow -\sigma$. Since all circulations are equal, H is invariant under this relabeling, so $H(\sigma) = H(-\sigma)$ and $H'(0) = 0$.

2.3 Geometric description

The deformation parameter σ is related to the Möbius conjugacy classification. The Möbius transformation $f(z) = \lambda z$ with $\lambda = e^{\sigma + i\alpha}$ generates orbits that are: logarithmic spirals for $\sigma > 0$, circles for $\sigma = 0$, and radial rays for $\alpha = 0$. The locus $\sigma = 0$ ($|\lambda| = 1$) is where spirals become circles; imposing $\mathbb{Z}_N$ symmetry from the azimuthal wavenumber converts the circles into N -gon meanders. This provides a useful geometric language for the spiral-to-polygon transition but is descriptive, not mechanistic.

3 Main Results

3.1 The energy maximum theorem

Theorem 1 (Energy maximum at the polygon). *For $N \geq 3$ vortices on a ring with pairwise interaction $h(r) = -\ln r$, the interaction energy $H(\sigma) = \sum_{j < k} h(|z_j(\sigma) - z_k(\sigma)|)$ along the centered spiral deformation (4) satisfies:*

(a) $H'(0) = 0$ (the N -gon is a critical point), and

$$(b) \quad H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0 \quad (\text{it is a local maximum}).$$

Proof. Part (a). The centered deformation (4) satisfies $H(\sigma) = H(-\sigma)$ by the symmetry $k \mapsto N - 1 - k$ (which sends the exponent $(k - (N - 1)/2)/N$ to its negative while preserving H by the equal-circulation assumption). Therefore $H'(0) = 0$.

Part (b). At $\sigma = 0$, the pairwise squared distance is $|z_j - z_k|^2 = 4R^2 \sin^2(\pi m/N)$ where $m = k - j$. Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ with $u = \sigma/N$. Then $\ln |z_j - z_k|^2 = 2uj + \ln R^2 + F(u, m)$. Differentiating twice:

$$F''(0, m) = \frac{m^2}{1 - \cos(2\pi m/N)} = \frac{m^2}{2 \sin^2(\pi m/N)}.$$

Since $d^2(2uj)/du^2 = 0$, only F contributes to H'' . There are $(N-m)$ pairs with separation m . Converting $d/d\sigma = (1/N) d/du$ gives (??). Every factor in the sum is strictly positive: $(N - m) > 0$, $m^2 > 0$, $\sin^2(\pi m/N) > 0$. Therefore $H''(0) < 0$. $\square$

3.2 The sign rule

Proposition 2 (Sign rule for pairwise interactions). *For N vortices on a ring with pairwise interaction $h(|z_j - z_k|)$: if h is decreasing ($h'(r) < 0$), then $H''(0) < 0$; if h is increasing ($h'(r) > 0$), then $H''(0) > 0$.*

This is verified computationally for nine interaction types ($-\ln r$, $\pm 1/r$, $\pm 1/r^2$, $\pm r^2$, r , e^{-r}) across $N = 3, \dots, 9$. All decreasing interactions give $H''(0) < 0$; all increasing give $H''(0) > 0$, with no exceptions.

Since the 2D Green's function $G(r) = -\ln r/(2\pi)$ is decreasing, the vortex interaction on any 2D surface is decreasing, and $H''(0) < 0$ is universal for 2D vortex systems. This would not hold in 3D, where the Green's function $G(r) = -1/(4\pi r)$ gives $G'(r) = 1/(4\pi r^2) > 0$ —increasing—which would make the polygon an energy *minimum*, reversing the Onsager selection.

3.3 The Onsager–Kraichnan mechanism

In 2D turbulence, energy and enstrophy $Z = \int q^2 dA$ are independently conserved. The Kraichnan (1967) dual cascade transfers energy to large scales (inverse cascade) and enstrophy to small scales (direct cascade, dissipated by viscosity). This drives the large-scale flow toward a state that maximizes energy at decreasing enstrophy—the negative-temperature regime of Onsager's (1949) vortex statistics.

At negative temperature ($\beta < 0$), the Boltzmann weight $P \propto e^{-\beta H}$ assigns higher probability to higher-energy states. Since the regular N -gon is an energy maximum along the spiral deformation (Theorem 1), it is a candidate for the most probable macrostate. A full characterization of H in the $(2N - 3)$ -dimensional configuration space—determining whether the N -gon is a global maximum, a saddle, or merely a directional maximum—remains open. The thermostat enabling Onsager relaxation is the enstrophy dissipation at small scales—a consequence of the 2D cascade, not an external assumption.

3.4 Eigenvalue branch disconnection

Within the WKB approximation, the Rayleigh–Kuo eigenvalue equation $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$ has real potential V , so the WKB exponent satisfies $s^2 \in \mathbb{R}$. Writing $s = \sigma + i\alpha_\rho$: $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$. When $V < 0$ (verified for Saturn: $V = -364$ at the jet center), $\alpha_\rho \neq 0$ and $\sigma = 0$. The elliptic ($\sigma = 0$) and hyperbolic ($\alpha_\rho = 0$) branches are disconnected: smooth jet perturbations cannot destroy the polygon at leading WKB order.

3.5 Thomson critical ratio

Proposition 3. *For $N = 6$ vortices on the unit circle with central vortex κ_0 , the 12×12 stability matrix has characteristic polynomial*

$$p(\lambda) \propto \lambda^2 \cdot (9 + 36\kappa_0/\kappa + 16\pi^2\lambda^2) \cdot Q^2,$$

giving exact critical ratio $\kappa_0/\kappa = -1/4$ (from symbolic computation, verified numerically). For $N = 3, 4, 5$: $\kappa_{\text{crit}} = -1/2$.

The jump at $N = 6$ shows the hexagonal ring has enhanced sensitivity to anticyclonic central perturbations. For Jupiter: $N = 8$ requires $\kappa_0/\kappa > 0.51$ (strong central cyclone needed); $N = 5$ requires only $\kappa_0/\kappa > -0.50$.

4 Planetary Verification

4.1 Measuring polygon quality: σ_{geom}

For N features at positions z_k , define

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N \left| \ln |z_k / z_k^{\text{ideal}}| \right|, \quad (5)$$

where z_k^{ideal} are the best-fit regular N -gon vertices. $\sigma_{\text{geom}} = 0$ for a perfect polygon.

4.2 Cross-planetary comparison

Table 1: Planetary polygon configurations.

System	N	σ_{geom}	σ_{flow}	Regime	Entry
Saturn hexagon	6	0.50	0.10	Close / Moderate	Rossby
Jupiter north	8	0.06	—	Close	Thomson
Jupiter south	5	0.05	—	Close	Thomson

All three configurations sit near $\sigma = 0$. Different dynamical mechanisms (Rossby stationarity for Saturn, vortex self-organization for Jupiter) arrive at the same geometric destination—the polygon locus—through different “doors.”

Saturn’s $\sigma_{\text{geom}} = 0.50$ is larger than Jupiter’s $\sigma_{\text{geom}} < 0.07$ because Saturn’s hexagon is a jet meander with amplitude $\varepsilon \approx 0.11$, while Jupiter’s cyclone rings are discrete point-like vortices that closely approximate a regular polygon.

5 Discussion

5.1 The answer

Rotating fluids make polygons because:

1. they are *two-dimensional*: the 2D Green’s function is decreasing, forcing attractive vortex interaction;
2. they are *Hamiltonian on a symmetric domain*: the N -gon is a critical point of H (equivariant critical point theorem);
3. the attractive interaction makes the N -gon an energy *maximum along the spiral deformation* ($H'(0) = 0$, $H''(0) < 0$, proven for all $N \geq 3$);
4. the 2D inverse cascade drives the system to negative temperature, where energy maxima are statistically selected.

The polygon is inevitable for 2D rotating fluids. It would not occur in 3D, where the Green’s function is increasing and the polygon would be an energy minimum.

5.2 Limitations

The energy maximum $H'(0) = 0$, $H''(0) < 0$ is proven along the centered spiral deformation (4)—one direction in the $(2N - 3)$ -dimensional configuration space. The full Hessian of H at the N -gon is not characterized; the N -gon could be a saddle point rather than a true local maximum. Determining the Hessian signature on the constant-angular-momentum surface is an open problem. The Onsager equilibrium is invoked but not derived for the specific planetary context. The QGPV (continuous) and point vortex (discrete) descriptions are connected through the shared 2D domain and logarithmic interaction, but a rigorous bridge is not established. Cassini and Juno data are digitized from published figures. Three configurations from two planets is suggestive but not conclusive for universality; Uranus and Neptune observations would strengthen the claim.

5.3 What is novel

The individual components—Onsager condensation, Kraichnan cascades, Thomson stability, 2D Green’s functions—are established. What is new is the chain connecting them: the closed-form proof that $H''(0) < 0$ for all $N \geq 3$; the sign rule $h' < 0 \Rightarrow H'' < 0$ connecting Green’s function monotonicity to polygon selection; the argument that two-dimensionality forces the entire chain; and the unified Saturn–Jupiter framework showing both arise from the same mechanism.

References

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